

Jason (Jingxuan) Liu

Irvine, CA | jasonliu131060@outlook.com | (949) 617-9378

EDUCATION

University of California, Los Angeles (UCLA) | B.S. in Electrical Engineering | GPA: 3.97/4.00 | Expected Jun 2029

Completed Coursework: Signals and Systems, Probability and Statistics, Foundations of Computer Vision

Planned / In Progress: Circuit Theory I & II Honors, Logic Design of Digital Systems, Analog Electronics I, Principles of Feedback Control, Digital Signal Processing, Introduction to Machine Learning

PROFESSIONAL EXPERIENCE

Undergraduate Researcher (SURP) | UCLA Integrated Sensors Laboratory | Jun 2026 – Aug 2026

- Built a dual-camera edge-AI wearable for LLM-based scene summarization, coupling a Himax WiseEye 2 for continuous low-res detection with an ESP32-S3 and high-res sensor for BLE streaming to the Gemini API
- Trained and deployed an INT8-quantized YOLOv5n model (1.7M parameters) to a Himax WiseEye 2 AI processor, achieving 0.833 mAP₅₀ for poster detection using PyTorch and Open Images V7
- Calibrated cross-camera perspective via OpenCV homography, mapping detection bounding boxes to an OV5640 high-res sensor and tuning exposure, gain, and dynamic cropping to minimize BLE payload size
- Ported an 8 MHz SPI burst FIFO driver for an ArduCAM OV2640 to Zephyr RTOS with nRF Connect SDK to benchmark power trade-offs, demonstrating BLE consumed 0.37 J/image over an nRF52840 DK versus 2.41 J/image over ESP32 Wi-Fi

ENGINEERING EXPERIENCE

Powertrain Project Engineer | Bruin Racing – Supermileage Vehicle | Jun 2026 – Present

- Designed and simulated a 120 W, 200 kHz active-clamp forward converter in LTspice to step down the 48 V powertrain bus to a galvanically isolated 12 V low-voltage rail for an ultra-efficient hydrogen fuel-cell vehicle, maintaining < 0.8% output ripple (< 94 mV_{pp}) across eight 0.25–10 A operating points
- Modeled a custom PQ32/30 N97 transformer in SPICE with a 6:4 turns ratio and $L_m = 180 \mu\text{H}$ nominal magnetizing inductance, parameterizing winding leakage and retuning 45–420 ns switching dead times to characterize hard-switching behavior and MOSFET voltage stress
- Designed mixed-signal sensing front ends for converter protection using an STM32G474, specifying Kelvin-sensed shunts, unity-gain-buffered dividers, and NTC circuits for on-chip ADC sensing and comparator-driven fault shutdown
- Simulated a two-switch forward-converter alternative that clamps primary switch nodes to the 48 V input rail, eliminating active-clamp reset overvoltage and targeting compliance with the Shell Eco-marathon 60 V maximum-voltage requirement

Low-Voltage Subteam Member | Bruin Racing – Supermileage Vehicle | Sep 2025 – Jun 2026

- Diagnosed and resolved critical CAN bus signal integrity failures and harness line breakage on-site at the 2026 Shell Eco-marathon in Indianapolis, restoring differential signaling, enabling passage of most electrical technical inspection
- Designed an STM32F446RE rear-control board for torque, pressure, and acceleration sensing, rear-lighting control, and CAN communication with the low-voltage system of the Supermileage Vehicle using KiCad
- Evaluated Hall-effect, hydraulic pressure, and strain-gauge sensing topologies for brake-light activation, selecting and deploying a strain gauge on the pedal assembly for reliable real-time trigger response

Project Member | UCLA IEEE Micromouse | Sep 2025 – May 2026

- Designed and assembled a 2-layer STM32F205 PCB integrating dual H-bridge motor control, magnetic encoder feedback, four-direction IR wall sensing, power regulation, and debug interfaces using KiCad
- Configured STM32 timers for quadrature encoder capture and four-channel motor PWM, implementing a 1 kHz closed-loop PD controller for straight-line motion and 90-degree turns
- Implemented time-multiplexed IR sensing with MOSFET-switched emitters, 12-bit ADC acquisition, and DMA averaging across 128 samples per reading to improve wall-detection stability
- Developed a 16×16 flood-fill maze solver in C using dynamic horizontal and vertical wall maps and breadth-first distance recomputation toward the four center goal cells

SKILLS

Prototyping & Test Equipment: PCB design, oscilloscope testing, soldering (hand, reflow and hot plate), simulation

Microcontrollers & Embedded Platforms: STM32, RP2040, ESP32, nRF52840, Arduino, Himax WiseEye 2

Protocols & Controls: I2C, SPI, BLE, PWM, PID Control, CAN

Software & Tools: STM32CubeIDE, ESP-IDF, nRF Connect SDK, OpenCV, KiCad, LTspice

Programming: C, C++, Python, MATLAB, R